

# AWS Networking

## Module 3 — AWS Technical Essentials




# Networking Basics

- **IP addresses:** unique identifiers for devices
- **Public IPs:** internet-routable
- **Private IPs:** internal only (10.x, 172.16.x, 192.168.x)
- **CIDR notation:** defines IP ranges (e.g. 10.0.0.0/16)

# 🌐 What Is Amazon VPC?

**Virtual Private Cloud** — your own isolated section of the AWS cloud.

- Logically isolated from other customers
- You define the IP address range (CIDR)
- Spans all AZs in a Region
- Full control over routing and security

# Subnets: Public vs Private

## Public Subnet

- Route to Internet Gateway
- Resources get public IPs
- Web servers, load balancers

## Private Subnet

- No direct internet route
- Private IPs only
- Databases, app servers

# 🚪 Gateways

## Internet Gateway (IGW)

- Connects VPC to internet
- Enables inbound + outbound
- Horizontally scaled, redundant

## NAT Gateway

- Outbound-only internet access
- For private subnet resources
- Managed, scales automatically

#  Route Tables

Rules that determine where network traffic is directed.

- Every subnet has exactly one route table
- Local route for VPC traffic (automatic)
- Add routes to IGW, NAT, peering, etc.
- Most specific route wins



# Security Groups

Virtual firewall at the **instance level**.

- **Stateful:** return traffic auto-allowed
- Allow rules only (no deny rules)
- Default: all outbound allowed, no inbound
- Can reference other security groups

# Network ACLs vs Security Groups

Firewall at the **subnet level**.

- **Stateless:** must allow both directions
- Allow and deny rules supported
- Rules evaluated by number (lowest first)
- Default NACL: allows all traffic

## Security Groups

- Instance level
- Stateful
- Allow rules only
- Primary defense layer

## Network ACLs

- Subnet level
- Stateless
- Allow + deny rules
- Additional defense layer

# 🛠️ Lab Time

## Create a VPC and Launch the App

### ⏱️ 25 minutes

- Build a VPC with public and private subnets
- Configure route tables and gateways
- Apply security groups for HTTP access